



Johann Schmidt

Research Scientist in Deep Learning

✉ johann.schmidt@ovgu.de
📍 Magdeburg, Germany
🌐 johann-schmidt.com
🌐 [linkedin.com/in/johann-schmidt](https://www.linkedin.com/in/johann-schmidt)
🐙 github.com/johSchm
🎓 Google Scholar

Research scientist in deep learning, working full time since 2020 on funded R&D projects with industry partners, from production scheduling to traffic signal control and acoustic traffic sensing. My PhD, pursued alongside, makes pretrained vision models robust to rotated and rescaled inputs, with papers at ICML, BMVC and ECML-PKDD. **I am looking for a research lead position in industry.** I will submit my thesis at the end of 2026 and am open to new roles from then on. I am a father of two, so a family-friendly employer matters to me.

12+

Years Coding Experience

18+

Theses Supervised

18+

Papers Reviewed

12+

Papers Published

EXPERIENCE

Research Scientist at OVGU · AILab, Otto von Guericke University Magdeburg, Germany 2020 – today
Full-time position on funded R&D projects with industry partners, with the PhD pursued alongside. The projects set the topics, which is why my work spans several fields.

- **KIVA-NET** · ZIM-funded project with Thorsis Technologies GmbH 2025 – 2027
Low-cost acoustic sensor that counts and classifies vehicles from sound alone, in real time on edge hardware. A video tracker labels the audio automatically. First models detect passing vehicles with 96.5% accuracy.
- **PASCAL** · BMBF-funded project with Thorsis Technologies GmbH 2022 – 2025
Traffic signal control on real-time V2X data for Magdeburg's urban testbed. Developed TransferLight together with a graduate student: a single reinforcement learning policy on a hierarchical graph neural network that controls any intersection layout and transfers zero-shot from synthetic to real road networks. It outperforms heuristic and learned baselines on travel time, waiting time and CO₂ emissions. Integrated with Thorsis into an operator-assistance prototype on edge hardware.
- **AI-Engineering** · BMBF-funded project of five universities in Saxony-Anhalt 2022
New Bachelor's program that combines AI with engineering, built by five universities and running since 2023. Contributed to the curriculum and led the development of the platform for sharing open educational resources across the universities.
- **SENECA** · BMBF-funded project with Thorsis Technologies GmbH and TECTRON GmbH 2020 – 2022
Real-time decision support for production scheduling at the electronics manufacturer Tectron, an NP-hard problem with up to 200 open orders per planning run. Developed deep learning approaches that find solution candidates with larger gains than existing heuristics. In Tectron's production, the final system beat manual planning on late orders and total tardiness.

EDUCATION

PhD in Deep Learning at OVGU · Otto von Guericke University Magdeburg, Germany 2022 – 2026
Thesis research since 2022, alongside the full-time project work. Submission at the end of 2026; the PhD ends in April 2027.

- **PhD Thesis** · AILab, Otto von Guericke University Magdeburg, Germany
Robust and Efficient Discriminative Deep Learning by Canonicalizing and Lifting Symmetry-Degenerated Signals. A small module in front of a frozen pretrained model corrects the view of each input to ease the downstream task, which adds robustness across views. The methods act as preprocessing add-ons to frozen off-the-shelf models.
- **OxML Summer School** · Oxford, United Kingdom August 2025

Sustainable Supply Chain DeepHack · 2nd place, Transatlantic AI Hackathon 2022
Built in one weekend with a team: automatic parcel volume estimation with joint planning of vehicle loading and route.

Master: Digital Engineering · Otto von Guericke University Magdeburg, Germany 2018 – 2020
Grade 1.4. Thesis at Fraunhofer IFF on vision-based human action recognition for human-machine interaction in industry, grade 1.1, published at ETFA 2021.

Bachelor: Electrical Engineering · Magdeburg-Stendal University of Applied Sciences, Germany 2014 – 2018
Grade 1.7. Thesis at ifak Magdeburg on a GUI for an ultrasonic sensor system on an embedded controller in C, grade 1.2.

SELECTED PUBLICATIONS

ECML-PKDD 2026	Zero-Shot Test-Time Canonicalization using Out-of-Distribution Scoring D. Lindner, J. Schmidt, T. Siegl, M. Becker, S. Stober. Machine Learning and Knowledge Discovery in Databases.
BMVC 2026	PPS: Plug-and-Play Saccadic Vision for Fine-Grained Classification J. Schmidt, S. Stober, J. Denzler, P. Bodesheim. British Machine Vision Conference.
ECCV: BISCUIT 2026	Geometrically Constrained and Token-Based Probabilistic Spatial Transformers J. Schmidt, T. Siegl, M. Becker, S. Stober. European Conference on Computer Vision: International Workshop on Biomedical Image and Signal Computing for Unbiasedness, Interpretability, and Trustworthiness.
AAAI: MALTA 2025	TransferLight: Zero-Shot Traffic Signal Control on any Road-Network J. Schmidt, F. Dreyer, S. A. Hashimi, S. Stober. Association for the Advancement of Artificial Intelligence: Multi-Agent Reinforcement Learning for Transportation Autonomy Workshop.
ICML 2024	Tilt your Head: Activating the Hidden Spatial-Invariance of Classifiers J. Schmidt, S. Stober. International Conference on Machine Learning (ICML).
HICSS 2024	Reviving Simulated Annealing: Lifting its Degeneracies for Real-Time Job Scheduling J. Schmidt, B. Köhler, H. Borstell. Hawaii International Conference on System Sciences (HICSS).
ESWA 2021	NeuroEvolution of augmenting topologies for solving a two-stage hybrid flow shop scheduling problem: A comparison of different solution strategies S. Lang, T. Reggelen, J. Schmidt, M. Müller, A. Nahhas. Expert Systems with Applications.

SKILLS

Foundation Models	Swin, SAM, EVA, ViT, ConvNeXtV2 (vision); CLIP (vision and text); EnCodec (audio); CLAP (audio and text)
LLM Tooling	llama.cpp for local models, Claude Code with MCP, agents and skills
Deep Learning Stack	PyTorch (Lightning, Geometric), Hugging Face, timm, OpenCV, Kornia, escnn; Weights & Biases, MLflow, Optuna, Hydra; NumPy, pandas, scikit-learn; TensorFlow in earlier projects
Software Engineering and Backend	Git; FastAPI, Flask; PostgreSQL, MySQL
Multi-GPU Training on Linux	Multi-GPU cluster with distributed training (Hugging Face Accelerate); TensorRT and edge hardware for deployment
Programming Languages	Python (main language), C / C++, Java, C#, JavaScript
Spoken Languages	German (native), English (daily working language)

LEADERSHIP AND SERVICE

Reviewing	18 papers for ICLR 2027, NeurIPS 2026, ECCV 2026, ECCV: BISCUIT 2026, NLDL 2026.
Machine Learning Paper Club	Initiated the machine learning paper club at OVGU and moderated it with colleagues, every two weeks from 2022 to 2026.
Thesis Supervision	18 Bachelor's and Master's theses, many with partners such as Fraunhofer IFF, Trumpf and SaxonQ. They led to two publications (AAAI workshop 2025, ECML-PKDD 2026).

TALKS AND TEACHING

Conference Oral	ECML-PKDD (Main Research Track and QCDS Workshop), 2026; AAAI (MALTA Workshop), 2025; HICSS, 2024; IFAC INCOM, 2021
Invited Talk	Friedrich Schiller University Jena, 2025
Seminar	Deep Learning Project Seminar on traffic signal control with reinforcement learning and graph neural networks, designed and taught on my own (winter semester 2024/25)
Training	Science communication seminar, NaWik (2023); eleven teaching workshops, OVGU (2021)